

The relative position of planes in space

Parallel planes, the test that establishes them, and the four properties that follow immediately.

LESSON 25–26

2 × 40 MIN

GEOMETRY 10, §12

EXTENSION

LESSON CLOCK

8'

22'

26'

20'

18'

6'

The three cases	8 min
The test	22 min
Four properties	26 min
Applications	20 min
Practice	18 min
Homework	6 min

LEARNING OBJECTIVES

- List the three positions of two planes.
- State and apply the test for parallel planes.
- Prove that a third plane cuts two parallel planes in parallel lines.
- Use parallel planes to prove segments equal.

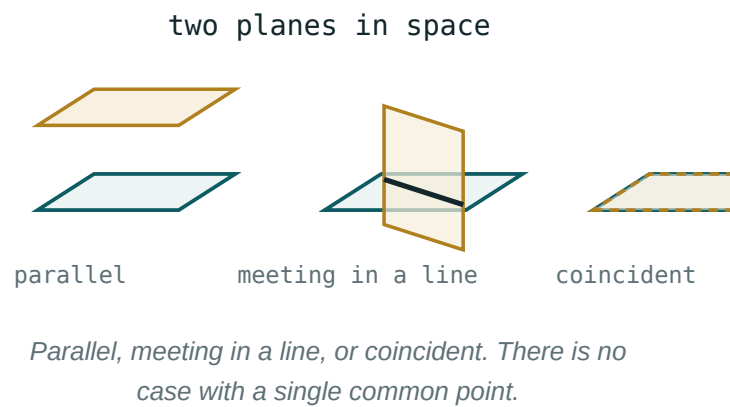
TEXTBOOK

National	pp. 113–124
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 12.1

01

Explanation

The three cases



Two planes with a common point share a whole line, by axiom S3. So either they have no common point (parallel), a common line (intersecting), or they are the same plane.

The test

TEST FOR PARALLEL PLANES

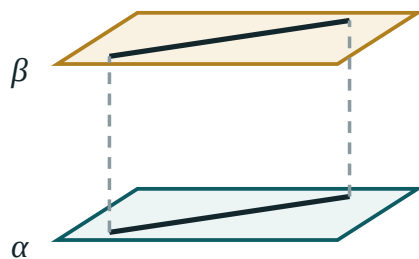
If **two intersecting** lines of one plane are respectively parallel to two lines of another plane, the planes are parallel.

TWO INTERSECTING LINES

One is not enough, and two parallel ones are not enough. A plane containing a single line parallel to β can still cut β — think of a page turning about that line. The two directions must be genuinely different to pin the plane down.

Proof sketch. If the planes met in a line c , then c would lie in α and be parallel to both of the given lines of α — impossible, since they intersect and a line cannot be parallel to two different directions in the same plane. ■

Four properties



$\alpha \parallel \beta$ — a third plane cuts them in parallel lines

A third plane cuts two parallel planes. The two lines of intersection are parallel.

1. A third plane cutting two parallel planes cuts them in **parallel lines**.
2. Segments of parallel lines cut off between two parallel planes are **equal**.
3. Through a point off a plane there is **exactly one** plane parallel to it.
4. If $\alpha \parallel \beta$ and $\beta \parallel \gamma$ then $\alpha \parallel \gamma$ — parallelism of planes is transitive.

Property 2 is the workhorse: it is why the lateral edges of a prism are all equal, and why the cross-sections of a prism parallel to the base are all congruent.

Applications

The prism. Its two bases lie in parallel planes. Property 1 makes corresponding base edges parallel; property 2 makes the lateral edges equal. Both facts are usually assumed — they are theorems, and this is their proof.

The midline of a tetrahedron. A plane through the midpoints of three edges is parallel to a face, because two of its intersecting lines are midlines, each parallel to an edge of that face.

Distance. Between two parallel planes the distance is the same everywhere — the length of any common perpendicular segment.

02 Worked examples

EXAMPLE 1 Prove that in a cube the planes $ABCD$ and $A_1B_1C_1D_1$ are parallel.

- ① $AB \parallel A_1B_1$ and $AD \parallel A_1D_1$.
Opposite sides of square faces.

- ② AB and AD intersect at A .
Two different directions.

- ③ Test applies.

ANSWER

$$ABCD \parallel A_1B_1C_1D_1$$

EXAMPLE 2 Two parallel planes are cut by a third in lines a and b . If a makes 35° with the line of intersection of the cutting plane and a fourth plane, what does b make?

- ① $a \parallel b$
Property 1.
- ② Angles with respectively parallel sides.
- ③ Equal or supplementary; here equal.

ANSWER

$$35^\circ$$

EXAMPLE 3 A tetrahedron $ABCD$. M , N , P are the midpoints of AB , AC , AD . Prove $MNP \parallel BCD$.

① MN is a midline of $\triangle ABC$, so $MN \parallel BC$.

② MP is a midline of $\triangle ABD$, so $MP \parallel BD$.

③ MN and MP intersect at M .

④ BC and BD lie in BCD .

Test applies.

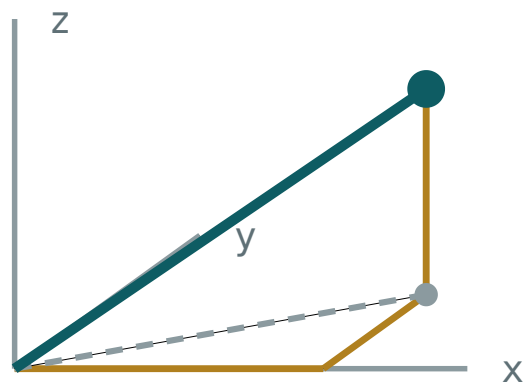
ANSWER

$MNP \parallel BCD$

03 Interactive model

Two sheets of card held parallel, cut by a third — the two cut lines are visibly parallel.

Two planes



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Coincident planes	Ustma-ust tushuvchi tekisliklar	Совпадающие плоскости
Test for parallel planes	Tekisliklar parallelligi alomati	Признак параллельности плоскостей
Intersecting planes	Kesishuvchi tekisliklar	Пересекающиеся плоскости
Line of intersection	Kesishish chizig'i	Линия пересечения
Distance between planes	Tekisliklar orasidagi masofa	Расстояние между плоскостями
Equal segments	Teng kesmalar	Равные отрезки
Transversal plane	Kesuvchi tekislik	Секущая плоскость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two distinct planes can share:

a point only

a line

two points only

nothing but a point

The test for parallel planes needs:

one parallel line

two intersecting parallel lines

three points

a perpendicular

A third plane cuts two parallel planes in:

perpendicular lines

parallel lines

one line

points

Segments of parallel lines between parallel planes are:

equal

proportional

perpendicular

unrelated

Through a point off a plane the number of parallel planes is:

0

1

2

infinitely many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the three positions of two planes

ANSWER parallel, intersecting, coincident

2. Can two planes meet in exactly one point?

ANSWER no

3. State the test for parallel planes

ANSWER Two intersecting lines of one parallel to two of the other

4. Are $ABCD$ and $A_1B_1C_1D_1$ parallel in a cube?

ANSWER yes

5. Are $ABCD$ and ABB_1A_1 parallel?

ANSWER no — they meet in AB

6. How many planes through a point are parallel to a given plane?

ANSWER 1

7. Is parallelism of planes transitive?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. Prove $ABCD \parallel A_1B_1C_1D_1$ in a cube

ANSWER $AB \parallel A_1B_1$ and $AD \parallel A_1D_1$, intersecting

2. Name a pair of parallel faces of a hexagonal prism

ANSWER the two hexagonal bases

3. Why is one parallel line not enough for the test?

ANSWER The plane can still turn about it

4. Distance between the base and the top of a cube of side 9

ANSWER 9

5. Tetrahedron: prove the midpoint plane $MNP \parallel BCD$

ANSWER Two midlines parallel to two edges of BCD

6. Two parallel planes cut by a third: the lines are

ANSWER parallel

7. Why are the lateral edges of a prism equal?

ANSWER Parallel segments between parallel planes

Hard · stretch and reason

7 PROBLEMS

1. Prove the test for parallel planes

ANSWER A common line would be parallel to two intersecting directions

2. Prove property 2 — segments between parallel planes are equal

ANSWER They form a parallelogram

3. Prove that if $\alpha \parallel \beta$ and a line meets α , it meets β

ANSWER Otherwise it would be parallel to both

4. A plane cuts a cube parallel to a face. Describe the section

ANSWER A square congruent to the face

5. A plane cuts a tetrahedron parallel to a face. Describe the section

ANSWER A triangle similar to that face

6. Prove: three parallel planes cut two transversals in proportional segments

ANSWER Reduce to the plane through the two transversals

7. Give two planes each parallel to the same line that are not parallel to each other

ANSWER Two walls of a room and a vertical line

07 Homework

Homework — 5 tasks

The test needs two *intersecting* lines. Say so in every proof that uses it.

1. Prove that in a cube the planes ABB_1A_1 and DCC_1D_1 are parallel.
2. Tetrahedron $ABCD$: prove the plane through the midpoints of AB , AC , AD is parallel to BCD .
3. Explain in three sentences why one parallel line is not enough for the test.
4. Prove that segments of parallel lines cut off between two parallel planes are equal.
5. Give an example of two planes parallel to the same line that intersect each other.